

ARCHIT SUREJA

SOFTWARE ENGINEER - II

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Senior Full-Stack Software Engineer with over 10+ years of experience specializing in high-scale distributed systems, Cloud Security, and Identity & Access Management (IAM). Proven track record at Amazon (AWS) leading the modernization of critical authentication frameworks, including the end-to-end delivery of Passkey MFA and IPv6 infrastructure. Expert in architecting secure, multi-region cloud solutions using Java, AWS CDK, and React/Next.js to drive operational excellence and business value.

SKILLS

LANGUAGE

Java, Kotlin, TypeScript, Swift, SQL

CLOUD & INFRASTRUCTURE

AWS (CDK, API Gateway, IAM, GovCloud), CI/CD (Jenkins, Sonar), Multi-region CDN Deployment, IPv6 Integration

BACKEND DEVELOPMENT

Spring, Node.js, MVC

MOBILE DEVELOPMENT

Android SDK, Jetpack Compose, Live data, Jetpack Navigation, Coroutines, MVVM/MVP, React Native, Redux, iOS (Xcode), Mobile SDK Development

FRONTEND

React, Next.js, CloudScape, HTML5/CSS3

TESTING & QUALITY

Cypress.io, Vitest, K6, Mockito, Mockk, TDD, E2E Canary Testing, Integration Test, Unit Test

ARCHITECTURE & TOOLS

Backbase, REST, Microservices, HLD/LLD, Git, Agile/Scrum, Room Database, SQLite.

WORK EXPERIENCE

AMAZON (AWS) | Toronto, ON

SOFTWARE DEV. ENGINEER - II | Oct 2021 - Cont.

Area of Focus: Identity & Access Management (IAM), Security, and Global Infrastructure

Backend & API Engineering

- Engineered scalable backend services using **Java and Spring Boot**, focusing on the end-to-end software development life cycle from technical design to deployment for **traditional (Password)** and **Federated (SAML 2.0, OIDC)** authentication flows.
- Designed and implemented secure, high-traffic **REST APIs** for MFA and identity workflows, enhancing API performance and diagnostic capabilities, utilizing industry-standard **encryption** for secure password transportation.
- Authored comprehensive **High-Level and Low-Level Designs (HLD/LLD)** for the migration of the Signup service and a major Sign-in Monolith into a microservices architecture, reducing deployment latency and technical debt.
- Developed and maintained robust backend components while managing **Git, CI/CD pipelines**, and automated service **Canaries** to ensure 99.99% availability.
- Architected secure cloud infrastructure using **AWS CDK**, specifically handling **API Gateway** integrations and multi-region service deployments.

Security & Core Infrastructure

- Implemented full **Dual-stack support** for the AWS Sign-In domain, including complex **threat modeling** and infrastructure-as-code deployments for **Classic and 100% compliance with GovCloud security mandates**.
- Spearheaded the backend integration for **Passkey as an MFA method**, resolving dual RPID challenges and enabling cross-platform authentication.
- Delivered critical Root account functionality (Login, Password Reset, Account Update) within the **EUSC partition**, ensuring resiliency against regional failures.
- Led high-severity incident resolutions and drove **security hardening** initiatives, such as protecting non-public paths and implementing **HTTP Slowloris protection**.

Full-Stack, Mobile & Testing

- Managed the complete software development life cycle, encompassing technical design, coding, testing, deployment, and maintenance.
- Developed **Android and iOS SDKs** for the AWS Sign-In service to provide secure, authenticated in-app browsing experiences.

AWARDS



EDUCATION

MASTER OF TECHNOLOGY

Major in Information and
Communication Technology

NIRMA UNIVERSITY

Ahmedabad, 2013

BACHELOR OF ENGINEERING

Major in Information Technology

GUJARAT UNIVERSITY

Ahmedabad, 2011

DIPLOMA

Major in Information Technology

TECHNICAL EXAM. BOARD

Gandhinagar, 2008

HOBBIES

Gaming, Cooking, Biking, Hiking

- **Optimized** frontend experiences across Desktop and Mobile by migrating the Sign-In service to a modern stack of **React, Next.js, and Node.js**.
- **Designed** comprehensive testing frameworks using **Cypress.io, Vitest, K6, and Mockito/Mockk**, maintaining over **integration tests and 100+ canaries** for Signin workflows.

Mentorship, Leadership & Operational Excellence

- **Spearheaded** strategic, cross-functional projects by collaborating with **8+** product and UX teams to deliver critical security features reaching **millions** of global users.
- **Mentored 1 intern and 3 junior engineers** on backend integration, security best practices, and achieving operational excellence.
- **Drove** team growth by actively participating in the **technical recruitment and interviewing** processes for engineering candidates.
- **Led** strategic, cross-functional team projects, collaborating with product stakeholders and UX teams to represent engineering considerations and meet milestones.
- Provided ongoing operational support to the team and identified opportunities to enhance system reliability and efficiency.

Hackathon Lead: ZKP-Based Authentication Framework

- **Designed** a **POC** authentication system utilizing **Zero-Knowledge Proofs (ZKP)** to allow users to authenticate without revealing underlying sensitive data.

INFOTREE TECHNOLOGY (EXPEDIA) | Pointe-Claire, QC

SOFTWARE ENGINEER (CONTRACT) | Sep 2021 – Oct 2021

Area of Focus: Geospatial Mobile SDK Development

- SDK development for map utils and features with use of **Google map** and **MapBox** libraries.
- Writing of Unit test cases, Instrumental test cases and automation UI test cases for robustness, security, usability.
- Consistently delivered and **implemented quality units** with functional features and **minimal defects**.
- Met product specifications and achieved goals for maintaining quality.

MOBIQUITY INC. | Ahmedabad, India

SENIOR ENGINEER | Oct 2017 – Jun 2021

Area of Focus: Digital Banking & Mobile Frameworks

Architecture & Mobile Development

- **Designed** complex technical architecture proposals to seamlessly integrate the **Backbase Framework** with various third-party libraries.
- **Programmed** robust mobile applications using **Android, React Native, and Kotlin**, utilizing object-oriented principles to enable code abstraction and high stability.
- **Improved** software quality and system reliability by implementing comprehensive integration tests, achieving **75%+ code coverage**.
- **Delivered** high-quality functional units with minimal defects, consistently meeting strict product specifications and stakeholder milestones.

Leadership & Mentorship

- **Supervised** a team of 3–6 engineers, providing technical guidance and resolving project obstacles through the application of company best practices.
- **Managed** project deliverable milestones for diverse stakeholders by ensuring immediate resolution of critical issues as they arose.
- **Mentored** team members on modern development standards and the adoption of emerging cloud services and ML trends.

Operational Excellence

- **Improved** software quality and reliability by **implementing integration tests** (with **75%+ code coverage**) for robustness, security, usability.
- **Configured** and optimized **Sonar and Jenkins CI/CD pipelines** to enhance code quality and accelerate build delivery speeds.
- **Orchestrated** development workflows within **Scrum and Agile** environments to maintain project velocity and transparency.

PRIOR PROFESSIONAL EXPERIENCE

SR. ANDROID DEV. | *e-Procurement Technologies* | Ahmedabad | Jan 2017 - Oct 2017

- **Developed** enterprise mobile applications using **Kotlin, Android SDK, and SQLite**, integrating advanced features like **Beacons and NFC**
- **Designed** reusable library modules and proof-of-concept applications to streamline integration with external systems.
- **Collaborated** with QA teams to execute end-to-end unit and post-production testing, ensuring high-quality, defect-free releases.

SR. ANDROID DEV. | *Techreco Solution* | Ahmedabad | Mar 2016 - Jan 2017

- **Programmed** mobile applications utilizing **Java and the Android framework**, delivering custom UI functional features with minimal defects.
- **Created** library modules and proof-of-concept apps to validate technical feasibility for new product requirements.

ANDROID DEV. | *Percept InfoTech* | Ahmedabad | Mar 2014 - Mar 2016

- **Developed** mobile applications and interactive experiences using **Android, Unity3D**.
- **Ensured** robust software delivery by maintaining a focus on functional quality and consistent unit performance.

PROJECT ENGINEER | *Institute for Plasma Research* | Gandhinagar | Jun 2012 - Mar 2014

- **Engineered** controlling system for prototype robot using **ROS (Robotic Operation System), Arduino, Raspberry Pi, Xbee Technology**.